

FIRST INTAKE RUNBOOK

Livestock Arrival & Envelope Commissioning

Prepared by: **Sipho Wiseman Khuzwayo**, Founder & Managing Director



The Intake Quality Gates

Operational instructions for Day-1 biological stocking across poultry and swine corridors.

Pre-Loading Preparation Steps



Biosecurity Wash

Complete multi-chemical foam wash across roofs, walls, and slats. Confirm wheel spray sumps are filled with disinfectant.



Thermal Warm-Up

Ignite the co-generation steam system **24 hours prior** to animal delivery to heat nursery floors to a comfortable 32°C.



Line Purification

Flush water lines with chlorine dosing. Confirm sediment disc and inline UV sterilizer units are running properly.

Intake Batch Sizing Math

5,760

Broiler Chicks Wave 1

Day-1 Capacity Load Targets

The first intake is sized conservatively to calibrate the biogas engine load and thermal exchange parameters smoothly:

- ✓ **Poultry Sprint:** 5,760 broiler chicks placed in the isolated brooding zone.
- ✓ **Swine Nursery:** 160 weaner pigs grouped across Pods 1 to 4.
- ✓ **Water Siphon Draw:** Sized for a starting fluid intake of 1,200 liters daily.

Broiler Chick Loading Protocol

Brooding Placement (Day 1)

Day-old chicks cannot self-regulate body heat. Minimize transit stress by initiating loading procedures quickly:

- ✓ **Zonal Isolation:** Confine placement within the partitioned 128 m² zone using thick PVC curtains.
- ✓ **Hydration Start:** Supply warm, glucose-treated water to boost energy reserves.
- ✓ **Crop Checks:** Physically check chick crop fullness 12 hours post-load to confirm feed access.



Nursery Pod Loading Protocol



Weaner Grouping & Slat Intro

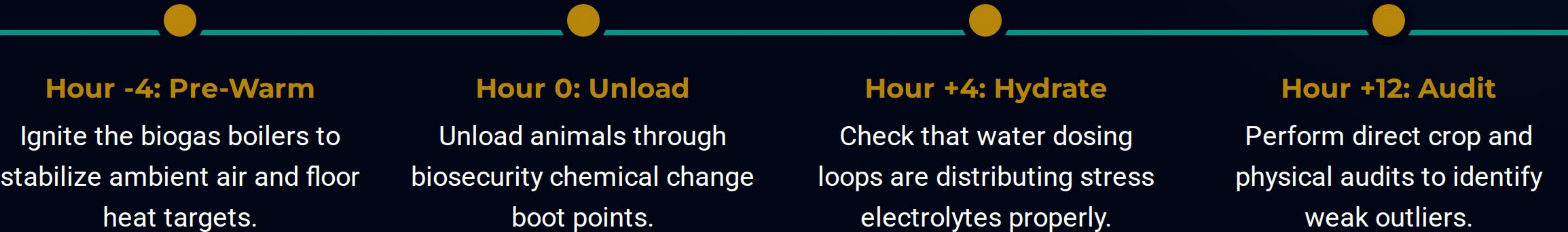
Transferring 7kg weaners into the elevated slatted nursery block must be managed to reduce pathogen introduction and sorting stress:

- ✓ **Weight Sorting:** Group pigs of uniform size into individual pods (Pods 1 to 4) to eliminate feeding fights.
- ✓ **Hygienic Slat Intro:** Place piglets directly on elevated virgin plastic slats over concrete flushes.
- ✓ **Thermal Check:** Verify steam-plumbed floor lines register 32°C at piglet height.

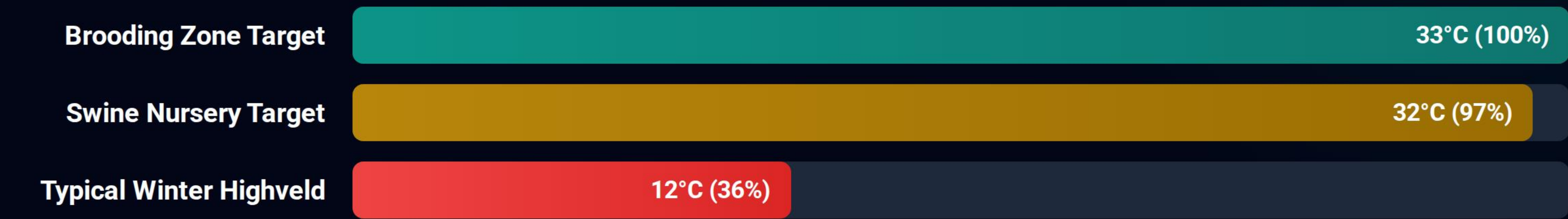
Day-1 to Day-7 Feed Schedule

Livestock Group	Day 1 - 2 Feed Class	Day 3 - 7 Feed Class	Primary Operator Action
Broiler Chicks	Pre-starter crumble on floor paper	Feeder tray transition with starter crumbs	Monitor chick distribution patterns across the heat zone.
Weaner Pigs	Pre-starter creep meal pellets	Grower phase pellet blend	Confirm creep feeders dispense dry meal without blocking.
Both Loops	Glucose-electrolyte water	Sanitized UV water lines	Test pipeline water pressure regularly to prevent dehydration.

The First 24-Hour Timeline



First Intake Climate Targets



By replacing heavy electrical heater grids with recycled engine exhaust floor steam, the target microclimates are maintained securely during cold Highveld weather transitions.

INTAKE MASTERED

Approved for Live Loading Ignition

Lwandile Engineering Projects (Pty) Ltd

Secunda & Kinross Agricultural Infrastructure Core | Just Energy Transition Implementation

Image Sources



<https://www.gov.mb.ca/agriculture/livestock/poultry/images/bba01s08d.jpg>

Source: www.gov.mb.ca



<https://image.made-in-china.com/365f3j00AJzHEptnhBuL/Pig-Farm-Farrowing-Box-Nursery-Stall-Grate-Pig-Slatted-Slatted-Floor-System-Plastic-Pig-Floor.webp>

Source: xinbaofeng88.en.made-in-china.com